

Computing the Singular Value Decomposition (SVD) with Fixed Point CORDIC Operations

Application to MIMO-OFDM

December 2006

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Table of Contents

1	Introduction	5
2	Brief Overview of MIMO OFDM Communication Receiver.....	5
3	Implementing the Pseudoinverse for Open Loop MIMO with an SVD of the H Matrix.....	6
3.1	Introduction.....	6
3.2	Derivation of SVD Based Computation of ZF Equalizer	6
4	Singular Value Decomposition of Arbitrary Complex 2x2 Matrices Using CORDIC Operations	7
5	Computing SVD of Rectangular Matrix with Square SVD Algorithms	10
6	Comparison of the CORDIC based Fixed Point Beamforming with Floating Point Beamforming MIMO OFDM.....	12
7	MIMO OFDM Repository.....	16
8	References	16
9	GNU Free Documentation License	17

List of Figures

Figure 1 2x3 MIMO Per Carrier Channel Model	6
Figure 2 Capsim® MIMO OFDM 2x2 Open Loop with Channel Modeling and Noise Specification..	13
Figure 3 Floating Point versus Fixed Point 2x2 Closed Loop MIMO OFDM High SNR.....	14
Figure 4 Floating Point versus Fixed Point 2x2 Closed Loop MIMO OFDM Medium SNR.....	15

1 Introduction

In this paper, the computation of the Singular Value Decomposition (SVD) of complex matrices will be presented. In particular, the use of CORDIC operations for fixed point implementations of the SVD of complex matrices will be introduced. This paper is an important introduction to the algorithms implemented in the GitHub repository for MIMO-OFDM:

<https://github.com/silicondsp/mimo-ofdm-release>

Table 1 MIMO OFDM Configurations

Configuration	Tx	Rx	CSI	FFTs	SVD	QR	Rate	Application
1x1 SISO	1	1	No	1	No	No	1x	Lowest Data Rate, Lowest Power
1x2 MRC	1	2	No	2	No	No	1x	Reliable Low Rate. Longer range more power.
2x2 Open Loop	2	2	No	2	2x2	No	2x	Medium Data Rate
2x2 Beam Forming	2	2	Yes	2	2x2	No	2x	Medium Date Rate more reliable then Open Loop
2x3 Open Loop	2	3	No	3	2x2	Yes	2x	Reliable Medium Data Rate
4x2 Beam Forming	4	2	Yes	2	2x2	No	2x	High Down link Data Rate, Reliable Up-link, Low Power Video
3x4 Open Loop	3	4	No	4	3x3	Yes	3x	High Data Rate
4x4 Beam Forming	4	4	Yes	4	4x4	No	4x	Very High Data Rate
1x4 Beam Forming	1	4	Yes	4	4x4	No	1x	Very Long Range

Notes: SISO Single Input Single Output, MRC Maximal Ratio Combining.

2 Brief Overview of MIMO OFDM Communication Receiver

A MIMO communication channel is shown in Figure 1. The Figure shows the complex frequency response for a single carrier in a MIMO OFDM system from each transmit antenna to each receive antenna. The complex elements h_{jk} ($j=1, 2, 3$) and ($k=1, 2$) form a complex matrix H . To implement a MIMO OFDM system, the receiver needs to estimate the channel per carrier in order to perform equalization and separate the multiplexed streams (or to generate a single stream in diversity schemes). To do this, each transmitted packet contains high throughput training symbols. The receiver processes these training symbols and estimates the channel. In an open loop zero forcing MIMO system, in order to separate and equalize the received channels, the Pseudoinverse [Weisstein] of the channel is computed for reach carrier's H channel matrix.

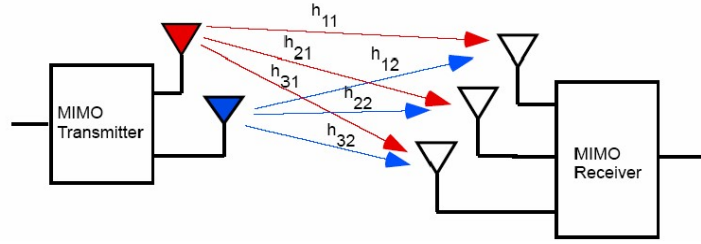


Figure 1 2x3 MIMO Per Carrier Channel Model

Thus, the matrix W is computed per carrier as follows:

$$W = H^+ = (H^\dagger H)^{-1} H^\dagger$$

The vector $\mathbf{y}$ contains the received complex carrier per receive antenna chain. Multiplying W by $\mathbf{y}$ yields the estimated transmitted complex carrier. In a 2x2 system, in effect, the two streams which were mixed in the multipath fading channel are now separated and can be demodulated independently and combined to get twice the throughput of a Single Input Single Output System (SISO).

$$\hat{\mathbf{x}} = W\mathbf{y} = \mathbf{x} + W\mathbf{n}$$

Where $\mathbf{x}$ is the vector of transmitted streams and $\mathbf{n}$ is the frequency domain additive white Gaussian noise.

3 Implementing the Pseudoinverse for Open Loop MIMO with an SVD of the H Matrix

3.1 Introduction

We will show that a basic 2x2 MIMO open loop equalizer can be derived from a basic 2x2 Singular Value Decomposition (SVD) of the H matrix. That is, an open loop MIMO system can be implemented with SVD. The key benefit is that for beamforming we use SVD and the SVD computation engine can be used for Open Loop systems.

3.2 Derivation of SVD Based Computation of ZF Equalizer

The MIMO Zero forcing equalizer is computed from the estimated MIMO channel using the Pseudo Inverse.

$$W = H^+ = (H^\dagger H)^{-1} H^\dagger$$

The streams are equalized (for example in high throughput separated) from the received streams $\mathbf{y}$.

$$\hat{\mathbf{x}} = W\mathbf{y} = \mathbf{x} + W\mathbf{n}$$

Compute the SVD of H and substitute for H.

$$H = U\Sigma V^\dagger$$

$$\begin{aligned}
W &= (Q^\dagger)^{-1} Q^{-1} Q U^\dagger \\
W &= H^+ = (V \Sigma^\dagger \Sigma V^\dagger)^{-1} V \Sigma^\dagger U^\dagger & (AB)^{-1} &= B^{-1} A^{-1} \\
W &= (V \Sigma^\dagger \Sigma V^\dagger)^{-1} V \Sigma^\dagger U^\dagger & W &= (\Sigma V^\dagger)^{-1} U^\dagger = (V^\dagger)^{-1} \Sigma^{-1} U^\dagger \\
Q &= V \Sigma^\dagger & V^\dagger V &= I \\
W &= (Q Q^\dagger)^{-1} Q U^\dagger & W &= V \Sigma^{-1} U^\dagger
\end{aligned}$$

So we can express the Pseudoinverse W as:

$$W = V \Sigma^{-1} U^\dagger$$

So, using an SVD 2x2 kernel we can support both beamforming and open loop MIMO equalization.

Note that Maximal Ratio Combining (MRC) is a 1x2 MIMO system and SISO is a 1x1 MIMO system.

4 Singular Value Decomposition of Arbitrary Complex 2x2 Matrices Using CORDIC Operations

This section contains the formulae and matrix manipulations for the Singular Value Decomposition of arbitrary complex matrices. The key point is to compute the SVD such that CORDIC computation units, rotations and arc tangents, can be used. The method is based on the work outlined in:

Nariankadu D. Hemkumar, "Efficient VLSI Architectures for Matrix Factorization,"
Ph.D. Dissertation, Rice University, April 1994.

A 2x2 arbitrary complex matrix is used to illustrate the technique.

Define the Matrix A ,

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = U \Sigma V^\dagger \quad (1)$$

We start out with the Matrix A and use *Octave* or *Mathematica* to compute the SVD:

$$A = \begin{pmatrix} 2 + 3i & 5 + 7i \\ 9 - 3i & 8 + 6i \end{pmatrix} = U \Sigma V^\dagger$$

Where, $\Sigma = [18.5769; 5.8221]$ and,

$$U = \begin{pmatrix} -0.77513 - 0.22045i & -0.30045 + 0.51021i \\ -0.51724 + 0.28818i & -0.20187 - 0.78017i \end{pmatrix}$$

$$V = \begin{pmatrix} -0.71051 + 0i & 0.70369 + 0i \\ -0.23065 + 0.66482i & -0.23288 + 0.67126i \end{pmatrix}$$

The first step in computing the SVD is to convert the matrix A into Polar form. This is accomplished using the ArcTan CORDIC function. Note that the ArcTan CORDIC function also computed the modulus. In "C" notation:

```
 $\theta = \text{ArcTanCordic}(x, y, \&r);$ 
```

$$\begin{aligned}
r_{11} &= \text{Abs}[a_{11}] \\
\theta_{11} &= \text{Arg}[a_{11}] \\
r_{12} &= \text{Abs}[a_{12}] \\
\theta_{12} &= \text{Arg}[a_{12}] \\
r_{21} &= \text{Abs}[a_{21}] \\
\theta_{21} &= \text{Arg}[a_{21}] \\
r_{22} &= \text{Abs}[a_{22}] \\
\theta_{22} &= \text{Arg}[a_{22}]
\end{aligned}$$

$$AA = \begin{pmatrix} r_{11}e^{i\theta_{11}} & r_{12}e^{i\theta_{12}} \\ r_{21}e^{i\theta_{21}} & r_{22}e^{i\theta_{22}} \end{pmatrix}$$

Define the following angles:

$$\alpha = \frac{\text{Imag}[a_{22}] + \text{Imag}[a_{21}]}{2}$$

$$\beta = \alpha$$

$$\eta = \frac{\text{Imag}[a_{22}] - \text{Imag}[a_{21}]}{2}$$

$$\omega = -\eta$$

Definition A Q transformation is a two-sided unitary 2x2 matrix transformation, where U_1 and V_1 , the left and right matrices, respectively, are given by:

$$U_1 = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{i\beta} \end{pmatrix}$$

$$V_1 = \begin{pmatrix} e^{i\eta} & 0 \\ 0 & e^{i\omega} \end{pmatrix} \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}$$

$$\omega = -\eta$$

The first steps are to use the U_1 and V_1 matrices to transform the matrix A into an upper triangle matrix R_L , where,

$$R_{lower} = RL = U_1 \times A \times V_1$$

Convert the Upper Triangular Matrix R_L to Polar Coordinates:

$$\begin{aligned}
r_{11} &= \text{Abs}[rl_{11}] \\
\theta_{11} &= \text{Arg}[rl_{11}] \\
r_{12} &= \text{Abs}[rl_{12}] \\
\theta_{12} &= \text{Arg}[rl_{12}] \\
r_{21} &= \text{Abs}[rl_{21}] \\
\theta_{21} &= \text{Arg}[rl_{21}] \\
r_{22} &= \text{Abs}[rl_{22}] \\
\theta_{22} &= \text{Arg}[rl_{22}]
\end{aligned}$$

Now we need to transform R_L into a real matrix R . We define the angles,

$$\alpha = \frac{\theta_{11} + \theta_{12}}{2}$$

$$\eta = \frac{\theta_{11} - \theta_{12}}{2}$$

$$\beta = \eta$$

$$\omega = -\eta$$

Define,

$$\phi p \psi = -\text{ArcTan}(r_{12}, r_{22} + r_{11})$$

$$\phi m \psi = \text{ArcTan}(r_{12}, r_{22} + r_{11})$$

$$\phi = \frac{\phi p \psi + \phi m \psi}{2}$$

$$\psi = \frac{\phi p \psi - \phi m \psi}{2}$$

$$U_2 = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{i\beta} \end{pmatrix}$$

$$V_2 = \begin{pmatrix} e^{i\eta} & 0 \\ 0 & e^{i\omega} \end{pmatrix} \begin{pmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{pmatrix}$$

Compute:

$$R = U_2 \times RL \times V_2$$

The Matrix R is a real matrix.

The next step is to use Jacobi Rotation to diagonalize R to obtain the Singular Values.

For Jacobi Rotations see: James W. Demmel, "Applied Numerical Linear Algebra", 1997 p. 249 Algorithm 5.12.

We have,

$$R = \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix}$$

Let,

$$x = r_{11} - r_{22}$$

$$y = 2r_{12}$$

In "C" notation compute θ :

```
 $\theta = \text{ArcTanCordic}(x, y, \&r);$ 
```

Now compute $\cos(\theta)$ and $\sin(\theta)$ using the CORIC Rotation module. In "C" notation, with $x = 1; y = 0$,

```
 $\text{CordicRotate}(\&x, \&y, \theta);$ 
```

This function returns, $x = \cos(\theta)$ and $y = \sin(\theta)$.

$$\Sigma = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Computation of U Matrix:

$$U = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \times U_2 \times U_1$$

Computation of V Matrix:

$$V = V_1 \times V_2 \times \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

5 Computing SVD of Rectangular Matrix with Square SVD Algorithms

The following is from [Delosme, 1991] and is concerned with computing the SVD of a rectangular matrix using the SVD of a square matrix.

Assume that we are interested in the SVD of an $m \times n$ matrix R . Let $m \geq n$. If not, Transpose the matrix if $m < n$. The Rectangular matrix R is decomposed into the product $Q \times S$ of an $m \times n$ matrix. Q satisfying $Q^H Q = I_n$ S is square of order n . Then the SVD of S is computed as:

$$S = U D V^H$$

$$\text{With } U^H U = V^H V = I_n$$

D is real Diagonal. The SVD of R ,

$$R = (Q U) D V^H$$

The implementation of the decomposition $R = QS$ is based on the Givens method in which plane rotations are applied to the rows of R in a specific order. For complex, an appropriate rotation is applied.

To illustrate we quote Example 3.6 provided in [Demmel, 1997]:

Example 3.6. We illustrate two intermediate steps in computing the QR decomposition of a 5-to-4 matrix using Givens rotations.

$$\begin{pmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & x & x \\ 0 & 0 & x & x \end{pmatrix}$$

to

$$\begin{pmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \\ 0 & 0 & 0 & x \end{pmatrix}$$

We multiply

$$\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & c & -s \\ & & s & c \end{pmatrix} \begin{pmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & x & x \\ 0 & 0 & x & x \end{pmatrix} = \begin{pmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & c' & -s' \\ & & s' & c' \\ & & & 1 \end{pmatrix} \begin{pmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{pmatrix} = \begin{pmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \\ 0 & 0 & 0 & x \end{pmatrix}$$

In the above, $c = \cos \theta$, $s = \sin \theta$ and $c' = \cos \phi$, $s' = \sin \phi$.

6 Comparison of the CORDIC based Fixed Point Beamforming with Floating Point Beamforming MIMO OFDM

We have implemented the 2x2 SVD (Beamforming) MIMO OFDM System using the Fixed Point CORDIC based 2x2 SVD kernel. In this section we compare the fixed point SVD implementation with the floating point SVD based on the LAPACK library. In the simulations the 2x2 Beamforming MIMO OFDM system shown in Figure 2 (using Capsim® and IEEE 802.11 54 Mbps streams) is used where the receiver is either based on the floating point LAPACK library, or the fixed point CORDIC based SVD. This comparison is key to exploring the degradation introduced by the fixed point implementation of the 2x2 SVD and to demonstrate that the fixed point 2x2 SVD performs well over a wide range of MIMO channels.

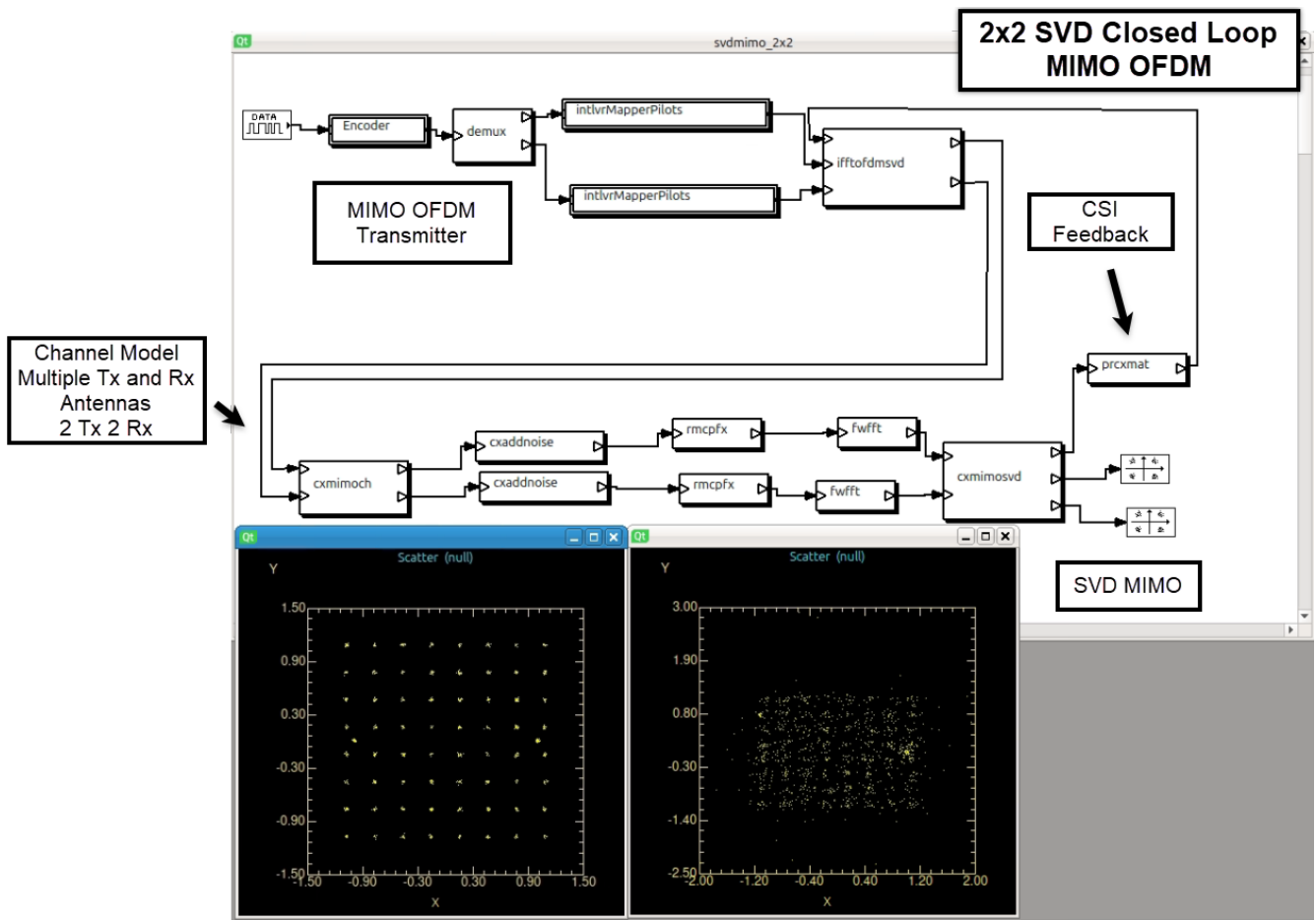


Figure 2 Capsim® MIMO OFDM 2x2 Open Loop with Channel Modeling and Noise Specification

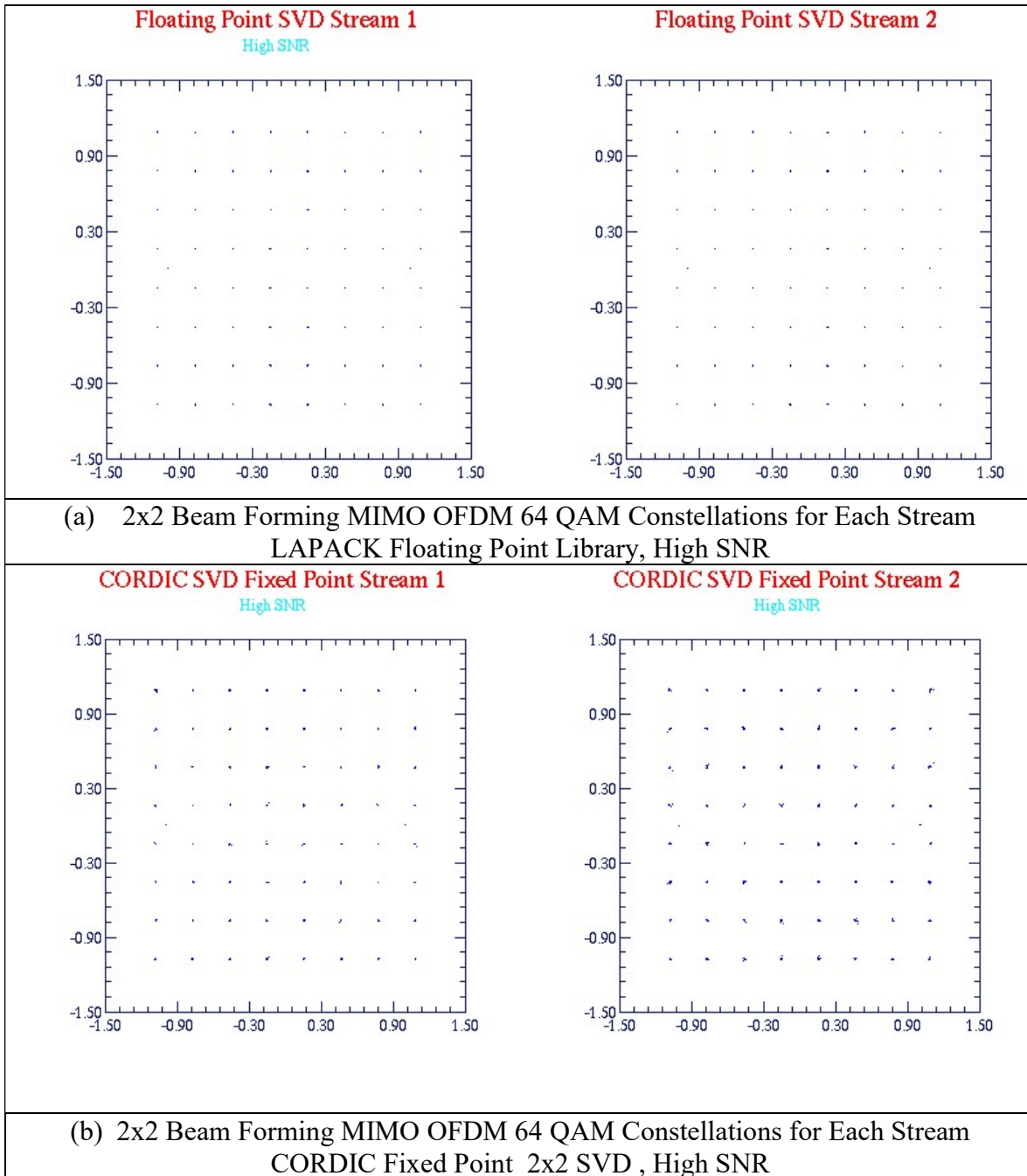


Figure 3 Floating Point versus Fixed Point 2x2 Closed Loop MIMO OFDM High SNR

As show in Figure 3, the fixed point CORDIC based 2x2 SVD MIMO OFDM system performs well compared to the floating point LAPACK based 2x2 SVD. The slight apparent degradation is due to the finite precision in the fixed point CORDIC 2x2 SVD implementation.

A key comparison is the case where channel noise is added. In this case, we expect that the finite precision fixed point CORDIC SVD will enhance noise and degrade performance compared to the floating point LAPACK implementation. This is shown in Figure 4 where noise variance of $1e-5$ (1×10^{-5}) was added to each receive chain.

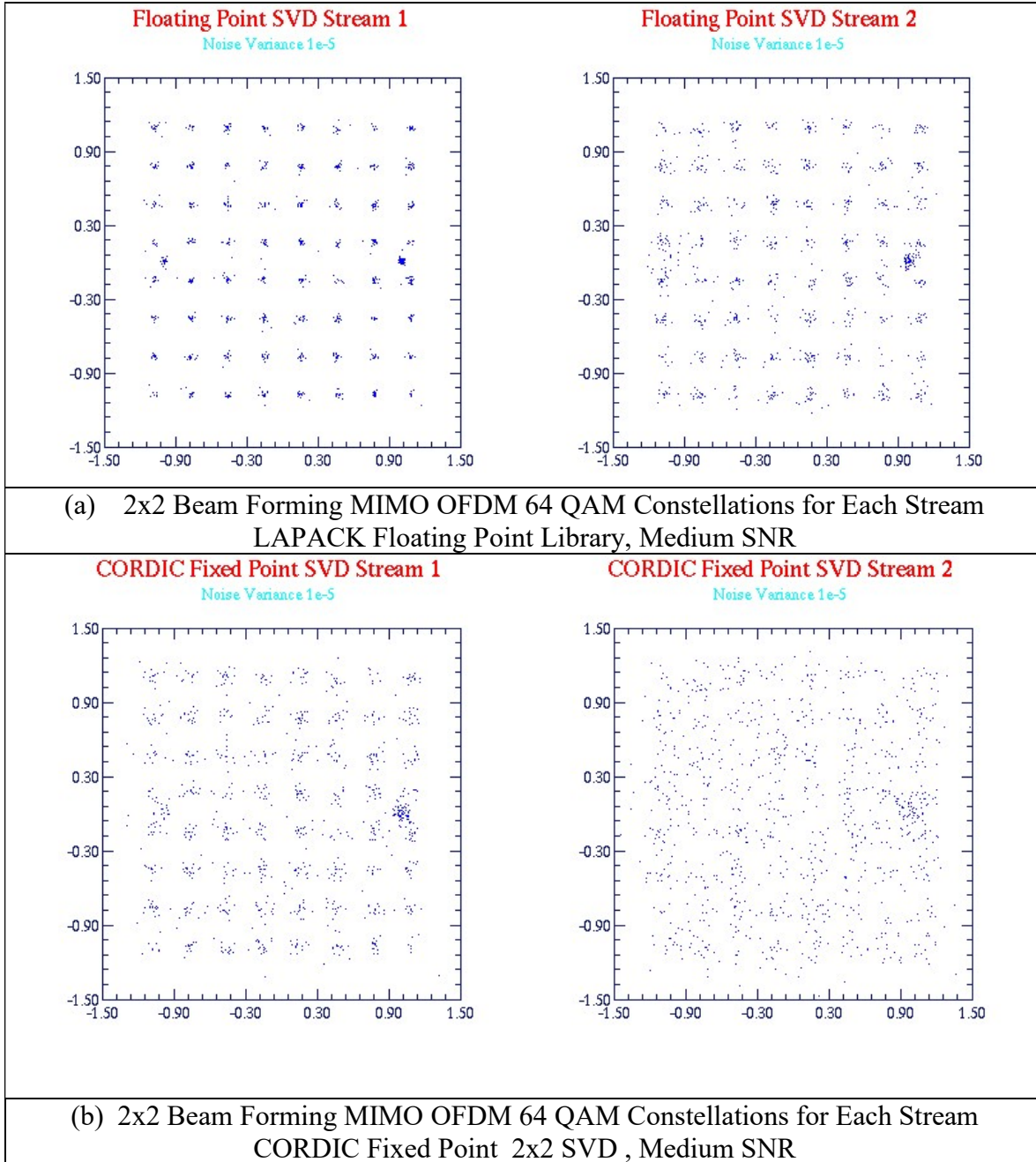


Figure 4 Floating Point versus Fixed Point 2x2 Closed Loop MIMO OFDM Medium SNR

7 MIMO OFDM Repository

A full implementation of the fixed point SVD computation in MIMO-OFDM systems is provided in the Capsim® based Block Diagram Modeling and Simulation environment at the GitHub Repository:

`https://github.com/silicondsp/mimo-ofdm-release`

The Capsim® simulation includes both Open Loop MIMO-OFDM systems and Closed Loop MIMO-OFDM systems with CSI feedback. Floating point implementations using LAPACK for matrix operations are also included for comparison with fixed point CORDIC operation implementations.

8 References

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<http://mathworld.wolfram.com/Pseudoinverse.html>

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